



Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院

ISE 333: User Interface Development (Design)

Chapter 9: Expressive Human and Command Languages

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Speech Recognition

- Conception
 - Speech recognition (or more generally natural language processing a.k.a. NLP for short) has made significant progress in recent years and is now well used



Speech Recognition

- Conception
 - Speech recognition (or more generally natural language processing a.k.a. NLP for short) has made significant progress in recent years and is now well used
- Place For Spoken Interaction
 - When users have physical impairments
 - When the speaker's hands are busy
 - Driving, cooking
 - When mobility is required
 - Photo self-taking
 - When the speaker's eyes are occupied
 - Shipping
 - When the user is unable to read or write
 - Children, very old generation, foreigners



Speech Recognition

- Speech Recognition Applications
 - Translation of speech to facilitate human communication
 - Foreign travelers or soldiers who must communicate in a language they do not know
 - Information voice response (IVR) systems, namely phone services, enable large financial savings for companies and provide 24/7 services for consumers
 - Production of text based on spoken input, a.k.a. dictation
 - Can be used to transcribe on-line speaking or recorded audio materials and allow users to compose a document
 - Ironically (in contrast with humans' listening), technical fields with a lot of jargon are good candidates for speech recognition because of the terminology's distinctive nature
 - Spoken commands for the UI
 - Users can speak commands that the UI is trained to recognize effectively
 - Speaker verification a.k.a. voice biometrics
 - While users answer questions, the system verifies that they are who they claim to be

Speech Recognition

- Designing Spoken Interaction

- Initiation

- The first step in using spoken interaction is for users to indicate that they wish to start the spoken interaction
 - Yet it is a tendency to have initiation-free spoken interaction which is more natural

- Knowing what to say

- Users need to know what can be said and reliably recognized

- Robust recognition

- Deal with regional or foreign accents and background noise
 - Deal with unknown new words
 - Users might misspeak or use the wrong terms
 - Match the semantic interpretation and contextual understanding that humans apply easily to predict and disambiguate what was said

Speech Recognition

- Designing Spoken Interaction

- Abnormality handling

- Identify novice users who attempt to use commands that are not recognized and switch them to a more directed mode that lists the possible commands

- Mapping to possible actions

- The secret of most successful speech recognition applications today is that they are limited to narrow application domains (so the world of actions is limited) and they use commands carefully chosen to increase recognition
 - You can also do in this way

- Feedback and dialogues

- During dictation or transcription, the recognized text is shown in the document being composed or in a dictation buffer, usually after a short delay (say one to two seconds)
 - The availability of a display can greatly speed up interaction by presenting the proposed action in detail and only asking users to confirm or cancel

Speech Production



- General Methods to Produce Speech
 - Canned speech
 - Consists of a fixed set of digitized speech segments which can be assembled together to create longer segments
 - “The next bus will arrive in” + “eleven” + “minutes”
 - The number of possible complete sentences is limited
 - The seams between segments may sound awkward
 - Formant synthesis
 - Produces entirely machine-generated speech using a set of algorithms based on phonetic representation of the text
 - Concatenated synthesis
 - Combines tiny recorded human speech segments into phonemes, words, phrases, and into full sentences
 - The voice is more natural but requires significantly more storage and computing power to assemble sentences on the fly

Traditional Command Languages

- Command-Line/Language Interface (CLI)

- Common commands

• ls	cd	pwd	mkdir	touch
• cp	mv	rm	cat	less
• head	tail	find	grep	ps
• kill	df	du	chmod	chown
• sudo	man	history	passwd	top
• sed	awk	ssh	scp	sshd

- CLIs are often preferred when the application is used in an advanced way

- Professionals use CLIs for hours every day
 - Web addresses or URLs (Uniform Resource Locators) can be seen as a form of command language